

A systematic review of shared personal informatics

Citation for published version (APA):

Xue, M., An, P., Guo, Z., Liang, R. H., Hu, J., Hansen, P., & Feijs, L. (2025). A systematic review of shared personal informatics. *Information and Software Technology*, 185, Article 107759.
<https://doi.org/10.1016/j.infsof.2025.107759>

Document license:

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DOI:

[10.1016/j.infsof.2025.107759](https://doi.org/10.1016/j.infsof.2025.107759)

Document status and date:

Published: 01/09/2025

Document Version:

Publisher's PDF, also known as Version of Record (includes final page, issue and volume numbers)

Please check the document version of this publication:

- A submitted manuscript is the version of the article upon submission and before peer-review. There can be important differences between the submitted version and the official published version of record. People interested in the research are advised to contact the author for the final version of the publication, or visit the DOI to the publisher's website.
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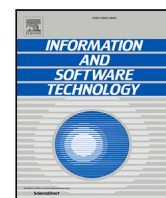
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Review article

A systematic review of shared personal informatics

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ARTICLE INFO

Keywords:

Personal informatics

Quantified self

Shared personal informatics

Quantified us

ABSTRACT

Personal informatics (PI) has gained great attention and become ubiquitous in people's everyday lives. Although an increasing number of studies set out to explore the social aspects of PI, there remains an unaddressed opportunity for a structured, systematic review to understand why and how the sharing happened, to inform future design and research. This systematic review summarizes the last 13 years of research on the diverse cases of shared PI practice from ACM, PubMed, and IEEE. 100 papers were analyzed, and four types of sharing were identified: Interpersonal targeting, Public broadcasting, Group monitoring, and Community exchanging. Notably, sharing extends beyond data exchange, evolving into a collaborative process across different PI stages. The review offers a taxonomy of shared PI practices and delineates design possibilities, facilitating future exploration in the field. Additionally, it identifies trends and patterns within existing work, suggesting design opportunities for future explorations of shared personal informatics.

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<https://doi.org/10.1016/j.infsof.2025.107759>

Received 1 August 2024; Received in revised form 15 April 2025; Accepted 16 April 2025

Available online 27 May 2025

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1. Introduction

Personal Informatics (PI), also known as quantified self, describes the process of people engaging in self-tracking activities for the purpose of gaining an understanding of oneself and achieving a healthier life [1]. The ubiquitous sensing technologies, such as smart wearable and mobile applications, have largely reduced the barrier for self-tracking. PI has grown out of the quantified-self group and got accepted by and integrated into the general public's daily life [2]. People track various aspects of their lives, such as physical activity, mood, finances, location, dietary intake, or anything that can be tracked and recorded using emerging sensors or apps. Self-tracking can sensitize people about multiple aspects of their daily life and thus support them to gain quantified understandings about themselves and make progress toward achieving a healthy lifestyle [3]. Moreover, in the clinical field, sensing technologies have been widely adopted by clinicians due to their accessibility and flexibility for patients' self-tracking [4]. The process of personal informatics practice was described in a classical model as a series of 5 stages: "preparation, collection, integration, reflection, and action" [1]. *Preparation*, where individuals determine their motivations and methods for collecting personal information; *Collection*, involving the observation and recording of personal data such as thoughts, behaviors, and interactions; *Integration*, where collected data is processed and transformed to enable reflection; *Reflection*, during which users explore and interpret their data, often through visualizations; and *Action*, where insights from the data inform decisions and behaviors to align with personal goals.

Recent researchers pointed out the trend of PI has transferred from "self-centric" into interpersonal contexts constantly [5] and showed the inherently social natures of some PI tools [6,7]. Sharing plays an increasingly significant and inevitable part of PI systems. People share their self-tracking information for various purposes, such as to gain social support [8], to create social comparison [9–11], to engage social sensemaking [4,12,13], or to promote behavior change [4]. As expected, concerns on privacy [14], data control [15], workplace surveillance [16,17], and commodification [18] had also been noticed. Collective PI practices are attracting an increasing amount of research endeavors. The self-experimenting members continued transferring from self-tracking and self-awareness into health collaboration communities like PatientsLikeMe [13], to make their self-discoveries "less anomalous, statistically significant, and scientifically rigorous" [19].

Previous work has shown great interest in addressing the social aspects of PI and has provided abundant cases for us to study with [20–24]. However, the focus of a majority of systems has been emphasizing on the sharing of data (e.g., visualization) [25–27] and the shared reflection [28–31] among multiple users. Nonetheless, in other stages of personal informatics [1] (i.e., preparation, collection, and action stage), a growing body of work spotted sharing behaviors and sharing effects as well. For instance, a family member may share the process of choosing the tracking tool and implementing the tracking with the actual user

who needs special care (shared preparation and collection) [32]; a group of runners collect their pace and distance data together during running (shared collection), which improved group performance and helped group members stay together (shared action) [33]. Beyond these contributions, several existing frameworks have explored social sharing in personal informatics mainly from a sharer-centric approach, such as individual-to-population-level transitions [5,34] and collaborative communities [19]. Moreover, current research has rarely systematically examined these diverse cases of sharing in personal informatics to inform and inspire future design or research. Therefore, an opportunity arises to systematically summarize current design cases to more comprehensively understand shared PI practices.

As such, we conducted a systematic literature review of current shared PI cases in the past thirteen years (2010–2023) to generalize and characterize the sharing practice of PI to support future designers and researchers. A systematic search was conducted in the ACM, PubMed, and IEEE databases, and the papers were selected based on our dedicated criteria, and summarized using inductive thematic analysis [35]. To generate design-oriented knowledge relevant to the domain of HCI (Human–Computer Interaction), we specifically targeted HCI due to its interdisciplinary focus on human behavior, technology, and design, making it a suitable domain for actionable insights that enhance user interaction with technology. This review addresses the following aspects: (1) Teasing out and classifying current shared PI practices across the classical PI stages [1]: context (preparation), collection, media (integration), and evaluation (reflection and action), to identify why and how sharing happens; (2) Identifying potential trends and design opportunities for future explorations of shared PI.

Overall, our study introduces a novel perspective by categorizing shared PI practices based on sharer-receiver network typologies. We identified four major types of sharing: *Interpersonal targeting*, *Public broadcasting*, *Group monitoring*, and *Community exchanging*. Then we characterized these shared PI cases addressing the classical stage-based model [1]. Besides the commonly addressed "shared data" (integration stage) and "shared sensemaking" (reflection stage), we argue that sharing could actually be carried out in all stages of a PI system. Our results in general show an increasing research interest in the past thirteen years of social sharing in personal informatics. Research interest in shared PI emerged and was explored in widespread populations and clinical fields beyond self-tracker communities. The sharing content covers wider facets of people's life (e.g., mental health and emotions) than before (e.g., physical activities, food). We found that current shared PI research places emphasis on small groups with relatively close relationships or online groups with community peers. We also see an opportunity in designing shared PI for co-located groups (e.g., co-workers, classmates, lecture audiences, etc.) under their specific co-experienced context. This review of the literature characterized various shared PI cases, evolved the definition of this class of tools, and addressed specific design opportunities for future shared PI research.

Table 1
The keywords cluster used in all databases.

Keywords cluster	Synonyms
“PI” Cluster	“personal informatics” OR “living by numbers” OR “quantified self” OR “self-tracking” OR “self-surveillance” OR “personal analytics” OR “quantified us” AND
“Share” Cluster	“share” OR “sharing” OR “group” OR “social” OR “interpersonal” OR “beyond individual” OR “organizational” OR “collective”

2. Method

2.1. Literature search strategy

To address the research question of why and how sharing occurs in shared PI cases, we followed Kitchenham’s systematic review process [36] to gather relevant cases within the field of HCI. We first conducted a preliminary search to identify existing systematic reviews and potentially related studies [36]. We assessed our review of interest in the field of design, computer science, engineering, and biomedical databases: Association for Computing Machinery (ACM), life science and biomedical (PubMed), and technical literature in engineering and technology (IEEE Xplore) were identified as our database.

Then we use various combinations of search terms derived from our research question [36]. In order to tease out articles regarding shared PI practices, two keywords clusters were defined: “PI” cluster and “Share” cluster. We identified “PI” cluster by extracting keywords from key references in this field [1,21,37–41], and we used the terms that informed in their papers as the searching synonyms of “PI” cluster. The “Share” cluster was identified based on synonyms, alternative spellings, and the authors’ knowledge of relevant publication headings [36]. Such as “shared” [28], “sharing” [42], “social” [20,23,24], “interpersonal” [5,43], “collective” [5]. Entries were included if the title or abstract contained the “PI” cluster AND the “Share” cluster (Table 1). The same query was used for every database. In addition to our search of three databases, we manually searched for relevant references within highly-regarded articles [5,21,43,44] that addressed shared PI, even if these articles were not indexed in the aforementioned databases. This approach was adopted to ensure the inclusion of crucial literature that might have been missed in the initial database search, particularly those with exceptional relevance and specific alignment with our review’s scope. In terms of time span, our review covers articles published from Jan. 2010 to Sep. 2023.

2.2. Article selection and extraction

According to the guidelines in making inclusion and exclusion criteria [45], we first defined what we mean by “shared PI” in our research question. We define shared PI as *sharing practices* in PI studies, instead of narrowing down the definition to a class of systems. Shared PI includes cases that embody sharing practices appears in any stage or stages of a PI (preparation, collection, integration, reflection, action). Previous works defined shared PI as “a class of tools that allows groups of users to collect, aggregate, analyze, and share personally relevant information” and name these tools as “interpersonal informatics” [43]. They emphasize on understanding self better by comparing with others in their social network, thus implying that the sharing is from multiple sharers, commonly among peers. To build upon this definition while also involve as many sharing contexts as possible, we include articles addressing a PI system with sharing function beyond sharing with peers. For example, there are cases including one sharer sharing data with his/her healthcare providers, which also fit the selection criteria of

Table 2
Article selection criteria.

Inclusion criteria	Exclusion criteria
(1) The article addresses a shared PI system OR people’s social PI practice;	(1) Works neither address a shared PI system nor include social PI practices;
(2) The use scenario of the system concerns at least two people;	(2) Sharing is not the main purpose/practice of the system;
(3) Sharing is (part of) the purpose of the system;	(3) Doctoral consortium, workshop, panel, opening keynote, courses, commentary;
(4) The article was published from Jan. 2010 to Sep. 2023;	(4) Works in progress and mock-ups;
(5) The article is written in English.	(5) Works include a shared PI but mainly reported in other expertise;
	(6) Systems that tracking non-human data;
	(7) Reviews;
	(8) Books;
	(9) Articles were not written in English.

shared PI practices in this review. The detailed article selection criteria are listed in Table 2.

An overview of the article selection process followed the PRISMA guidelines [46] and was described in Fig. 1. An overall of 666 articles were retrieved through the searches within the three databases. After screening the title, abstract and full-text, 575 articles were filtered out by our exclusion criteria. On top of that, 8 articles which are proposals or extended abstracts describing a system development in progress, were excluded because no related follow-up studies were found at the end of the database screening. 17 additional works were included, due to the aforementioned extra manual search based on the references of highly relevant articles. Articles describing the same shared PI system were combined and counted as one case in our list. Finally, 100 items met all the inclusion criteria and were included in this literature review. The data were extracted using inductive thematic analysis [35] to formulate the predefined categories of features among the selected articles, as listed in Fig. 2. After the first exclusion process, a cross review was performed by the second coder (name). Then the two coders settled the disagreements, summarized the characteristics, and refined the subcategories through a series of collaborative sessions.

3. Results

This section provides an overview of the selected papers and identifies four generic types of shared PI practices. It examines the contextual aspects of sharing and emerging design trends while analyzing data collection, media, and evaluation. Together, these analyses highlight key design aspects and research interests in the HCI community.

3.1. Overview of selected papers

An overview of the shared PI systems in the selected 100 articles are characterized and summarized. The detailed features of these systems are listed in Fig. 2 align with the publication year.

3.2. Four generic types of shared PI

First of all, we categorize these shared PI cases based on the typology of their sharer-receiver network (Fig. 3). Four types of sharing are identified: *Interpersonal targeting* (often one person to one person, or O2O), *Public broadcasting* (often one person to many persons, or O2M), *Group monitoring* (often many persons to one person, or M2O), *Community exchanging* (often many persons to many persons, or M2M). We define each type of sharing supported with typical cases as follows.

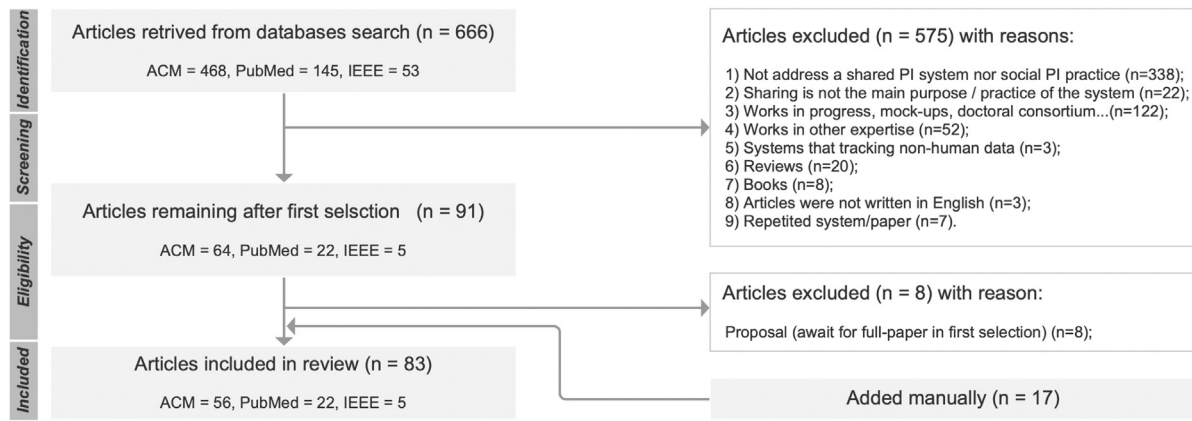


Fig. 1. The article selection flowchart.

- *Interpersonal targeting* (n = 25) often describes a sharing activity between two individuals with relatively close relationships. For example, a smoker shares his smoking habits (amount, trigger, time, location) with his counselor to discuss his quitting progress [30]. Another example is, a daughter helped her father, who is unfamiliar with new technologies, to track his steps and food [32].
- *Public broadcasting* (n = 10) often describes a mono-directional sharing activity without necessarily hoping others to share back. For example, a stroke patient continues updating his rehabilitation vlog and shares his experiences through a public YouTube channel [47]. Another example is, one captures his/her memories and feelings about a specific place using wearable accessories when cycling, and post them online through social networks [48].
- *Group monitoring* (n = 9) often describes passive or implicit sharing of relevant information from clients to professionals under the clients' consent. For example, a healthcare provider monitors multiple diabetes patients' medication adherence and blood glucose through cloud-based applications [49].
- *Community exchanging* (n = 56) often describes a reciprocal sharing activity within an ad-hoc community where people have similar goals or interests. For example, a group of primary school children share their physical activity data in their classroom through a shared display [14]. Another example is, users with mobility impairments share their activity data with peers who have similar impairments through the website PatientsLikeMe [13].

Several related frameworks have also characterized social sharing in personal informatics from different perspectives. Paton et al. presented a three-level framework to describe self-tracking information sharing that can develop from self-experimentation to group experiments, and ultimately to the entire population, which can enable sharing between health systems [34]. Swan sets a vision of the shift from “n = me” to “n = we”, as self-tracking individuals come together in collaboration communities [19]. Murnane et al. explored the individual patient's social relations in PI-supported treatment for bipolar disorder [5] based on the ecological system theory [50]. However, as different from these mentioned frameworks, in our work, we formulate the shared PI categories based on the typology of sharer-receiver networks (instead of a sharer-centric perspective). As a result, our categories reveal agreements with, as well as new insights in addition to these prior categories. For instance, the receivers of *Interpersonal targeting* (O2O) category are aligned with the receivers described to have a ‘microlayer relationship’ with the sharer in [5]. And the *Public broadcasting* (O2M) category has receivers that correspond to receivers who have an ‘exo-layer relationship’ with the sharer as described by [5]. Moreover, our categories also surface two distinct types of shared PI practices that are not explicitly differentiated by the prior frameworks: *Group monitoring*

(M2O), and *Community exchanging* (M2M). By differentiating these four types of shared PI practices that have rather different sharing cases, we hope to build on the prior works while also help designers to more concretely see the possibilities in the design of shared PI.

3.3. Contextual aspects of shared PI

To understand the context of shared PI practices, generalize patterns for specific user groups, and engage sharing in future shared PI design, we characterize the diverse shared PI cases from the following contextual aspects: *sharing purpose* (why people share), *sharer* (who is sharing), *receiver* (sharing to whom), *group size* (how many people are involved), *sharing content* (what content is being shared), and *sharing platform* (the tools for sharing). In this subsection, we first extract the contextual aspects of sharing, followed by comparing the four types of sharing addressing each aspect. Then, we map these contextual aspects of reviewed cases over time and summarize the trend.

3.3.1. Identify the contextual aspects

We identify six contextual aspects of these shared PI cases. The extracted keywords and descriptions are presented in Fig. 4. We further cluster the four types of sharing addressing each contextual aspect in detail (Fig. 5).

- *Sharing purpose*: We characterize the purposes that motivate people to share their self-tracking data to others into 6 sub-classes: exchange support (48%); get information (21%); comparison (9%); connectivity (10%); collaboration (11%); and reward (1%). Phrases in the articles like “support others in rehabilitation” [47], “receive and give support” [51], “community empowerment” [52] are categorized as “Exchange support”, because the sharing purpose is for giving and receiving social support. When people share their data for gaining answers, feedback, or comments from others, the purpose is categorized as the sub-class “Get information”. The purposes that describe a situation to “leverage the data of others to support the understanding of one's own data” [53], “to compare” [20,27,54], or “social sensemaking” [23] are classified under the “Comparison” sub-class. Sharing to improve “social connection” [55], “social connectedness” [56] are categorized as “Connectivity”. “Collaboration” describes the purpose of “support collaboration” [57], “collaborative sensemaking” [58], or “facilitate a collaborative process” [18]. The sharing in [59] is stimulated by monetary incentives, so the purpose is described as “Reward”. The sharing purpose of *Public broadcasting* is mainly for exchanging support (80%). On the contrary, *Group monitoring* mainly aims at getting information from experts and professionals (78%) (Fig. 5a). *Community exchanging* commonly aims to exchange support (61%) and social comparison (14%).

#	Ref	Users		Sharing Context		Data Collection			Feedback		Evaluation					
		What to share	Share to whom	Group size	Control	Benefit	I/C	Tool	Content	Ano Me	Format	Platform	No.	Duration	Qualitative	Quantitative
1	Paul et al. 2010	ED/Emergency Department care providers	Care providers	57, 142 residents, 34 nurses as ED charge nurse	System	Self	Others	commercial software (Edeyes)	each patient's ED number, physicians, residents, and nurses assigned, tests ordered, level of acuity, chief complaint, length of stay, and comments.	x	Self-Others (Patient-oriented information)	Shared display	17	SD 170 hours	Interview & Observation	N/A
2	Wicks et al. 2010	Patients	other patients like them in the same condition	forum community	User + System	Self	Others	commercial website	general questions and community-specific questions	y	Self-Others (Information-oriented)	Online platform (PatientLikeLife platform)	7000	-	N/A	Online logs, Site Use, Survey
3	Grimes et al. 2010	Generic users	the local neighborhood	residents community	User + System	Self	Others	commercial hardware (use cell phones, clips that detail their experiences)	audio recordings	y	Others	Recording clips	12	UD 4 weeks	Interview & Observation	N/A
4	Bales et al. 2011	Long-distance couples	partner	2	User + System	Self	Others	custom platform (smartphone Application)	locations	x	Others	Application	7*2=14	UD 2 weeks	Interview & Journal	Device logs
5	Luthi et al. 2011	older-ging experiences & friends, parents, other people's review	family/friends (4-6)	family/friends (4-6)	User + System	Self	Others	custom platform (smartphone Application)	bullet-going experiences, reviews from friends who, when, caption of submission (The post's health rating and their emotion are not visible).	x/y	Self-Others	Mobile Application	35	-	Reviews	Device logs
6	Baumer et al. 2012	health decisions	social network of friends (4-6) on VERA	7-11 peers in the same platform	User + System	Self	Others	custom software (mobilephone app)	goals, goal progress, and physical activities	y	Self-Others	Mobile Application	45+24=69	SD 2-4 weeks	Interview	N/A
7	Manson et al. 2012	physical activity-related updates	social network (facebook)	selective friends on facebook	User + System	Self	Others	custom software (mobile applications)	goals, goal progress, and physical activities	y	Self-Others	Social network(Facebook NewsFeed)	23	SD 4-week	Interview	N/A
8	Shvach et al. 2012	heart rate	partner and friends	2-7	System	Self	Others	commercial hardware (A heart-beat belt I Zephyr-Hold BT), a laptop(HP EliteBook 8440P)	heart rate	x	Self-Others	Shared display (laptops)	last 14+ 12-Hem ly 10-36	SD 2 weeks	Interview	N/A
9	Cutone et al. 2013	Location	social community membership	community	System	Self	Others	custom software (embedded smartphone sensors)	frequently visited locations, periodical movement patterns, and social interaction changeover time	y	Self-Others	Application	136	UD 1 month	N/A	Device logs, Survey
10	Epstein et al. 2013	fine-grained tap activity	friends, co-workers, siblings, and significant others	0-13 (Avg 4.33)	System	Self	Others	commercial hardware (FitBit®), commercial software (JavaToneUP2), and commercial software (mobile phone application 1)	steps	x	Self	Application	12	-	Interview	N/A
11	Malassa et al. 2013	memories	friend	friend or social network	System	Self	Others	commercial hardware (Liyad Arduino, I accelerometer sensor, temperature sensor, tri-color led, etc.), mobile phones and PC	photos, speed of travel, the temperature, etc.	x	Self-Others	Application	6	SD 2 days	Interview	N/A
12	Pelayo et al. 2013	feedback (personal interest) of the same lecture	other audiences in the same lecture	other audiences	User + System	Self	Others	custom software (smartphone application)	self-report interest of the lecture	y	Self-Others	Mobile Application	120	SD 1 month	N/A	Survey
13	Curti et al. 2013	heart rate	social network from family members to strangers	online community(8)	System	Self	Others	commercial hardware and software (Polar HR sensor, RunKeeper)	heart rate, average HR, pace, location, total heartbeats	x	Self	Mobile Application	52+9=61	-	Interview	Device logs, Survey
14	Maurolo et al. 2014	average pace, distance, and duration	running group (4-9)	running group (4-9)	System	Self	Others	commercial software; commercial hardware (Polar sensor); custom software	pace, duration, distance, and heart rate	x	Group	Shared display on wearable E-taille	52	SD 30 mins+	Interview & Observation	N/A
15	Hutton et al. 2014	location	A small subset of people	A small subset of people	System	Self	Others	custom software (Android-smartphone App)	location	x	Self-Others	Application	22	1 week	N/A	Device logs
16	Walsh et al. 2014	steps	social groups (friends)	friends	System	Self	Others	commercial hardware (Fitbit)	steps	x	Self-Others	Online platform	74	UD 7 days	N/A	Steps, Survey
17	Chang et al. 2014	food	colleagues	colleagues	User + System	Self	Others	public display + custom software (mobilephone application)	food (lunch log)	y	Aggregated ALL	Public display	171	UD 3 weeks	N/A	Device logs, Survey
18	Lingo et al. 2014	Activity, Sedentary Lifestyle, stress level, travel tracker	colleagues and managers (anonymous colleagues)	selective audience (a group of colleagues)	System	Self	Others	commercial hardware (wearable device) and HCM (Human Capital Management) data store	Activity, Sedentary Lifestyle, stress level, Sleep Quality, Correlating Factors with Sleep Quality, travel tracker	x/y	Self-Others	Online platform	19	SD 1 week	N/A	Survey
19	Parlier. 2014	the steps they are taking to eat healthily	the local neighborhood	residents community	User + System	Self	Others	SMS/MMS	photographs of foods they are eating, preparing, and purchasing, food establishments to pursue a nutritious diet	x	Aggregation ALL	Shared display	43	UD 6-week	Interview	N/A
20	Rod et al. 2015	physiological state	partner/housemate	2	System	Self	Others	I commercial hardware and software (Polar HR heart rate monitor and its app)	heart rate data of physical activity	x	Self-Others	Drinks	6	UD 2 weeks	Interview	N/A
21	Vyas et al. 2015	steps	group members, and global community	within team & global community	System	Self	Others	commercial hardware (wearable pedometer) & commercial website	steps	x	Self-Others+ World Average	Online platform	17	SD 45 mins	Interview	N/A
22	Snyder et al. 2015	mood	friend	2	System	Self	Others	commercial hardware (Personal Input Pod (Apple Watch HUE), smartphone (for android device))	skin conductivity (EDA sensors)	y	Aggregated ALL	Shared display (light)	22	-	Interview & Video Audio recording	N/A
23	Pay et al. 2015	smoking habits	counselors	2	User + System	Self	Others	custom software (mobilephone App)	amount, trigger, time, location	x	Self (Expert comments)	Application	13	SD 3 weeks	N/A	Device logs, Survey
24	Khorakhan et al. 2015	sleep and heart rate data	friends and professionals	2, a pair of 2 people	System	Self	Others	commercial software (Diapora OSMP and commercial software (Routa) and commercial hardware (Fitbit)	sleep and heart rate data	x	Self	Online platform	17	SD 3 weeks	Interview	Survey
25	Kuznetsov et al. 2015	DNA information	online community (peers in the same platform)	peers in the same platform	User + System	Self	Others	commercial split kit	saliva sample	x	Self-Others	Online platform	6	SD 3 month	Interview	N/A

(a) Part-1

Fig. 2. Selected papers.

#	Ref	Users	What to share	Share to whom	Group size	Control	Benefit Self Others 3rd Party	I/ C	Data Collection			Anno Me	Feedback		No.	Duration	Evaluation	
									Tool	Content	Vis		Format	Platform			Qualitative	Quantitative
26	Rocksbay et al. 2015	Generic users	physical activity	friends, relatives or coworkers, and strangers	1) 4, 5 teams of 4 people in each team 2) 2, 7, 7 teams	User + System	✓	✓	commercial hardware (accelerometers in iOS and Android devices)	a variety of physical activity (including shaking, jumping, and cycling)	x	✓	Self-Others+ Ranking	Mobile Application	20-31= 51	SD: 2 weeks+2 weeks	Interview	Device logs
27	Munson et al. 2015	Generic users	commitments to physical activity	social network	(~20 Facebook friends)	User + System	✓	-	commercial hardware (Fitbit) & custom software (website) & commercial software (Facebook)	steps	x	✓	Self	Mobile Application	81	SD: 12 weeks	N/A	Device logs, Questionnaires
28	Kim et al. 2016	Lifestyle diseases patients	food intake, activity, sleep data	medical experts	2	System	✓	✓	I custom software (mobile app) & commercial hardware (Mifit+Fitbit)	food intake, activity, sleep data	x	✓	Self	Online platform	20	8 weeks	Interview	N/A
29	Edelen et al. 2016	Single people	personal profile	dating partner	2	User	✓	✓	I self-chosen tools	personal informatics	x	✓	Others	F2F	11	SD: 3 hours	workshop events, interview	N/A
30	Kaulie et al. 2016	Generic users	steps and meals	intermediaries (family members)	2	User + System	✓	✓	I commercial software (pedometer app) and I custom website	steps and meals	x	✓	Self	Website	30	UD: 3 weeks	Interview	Device logs
31	Gorm et al. 2016	General employees	steps	other employees from other companies	participants in the campaign	User + System	✓	✓	I self-chosen step counter/App	steps	x	✓	Self-Others	Website	28	SD: 4 weeks	Interview & Observation	N/A
32	Berthel et al. 2016	Generic users	exchange life/ing streams	co-located people	2	User + System	✓	✓	commercial hardware (Microsoft SenseCam, Narrative Clip 1)	recording from co-located peers and any available capture devices	x	✓	Self-Others	System-wide URL	20	-	N/A	N/A
33	Accredited Social Health-Activists (ASHAs)	2016	visit clients (frequency)	peers (other ASHAs)	peers	User + System	✓	✓	I form + custom platform (web- and voice-based ASHA systems)	the number of clients that an ASHA visited	✓	✓	Self-AvgOthers	Application	142	SD: 12 months	Interview	Device logs
34	Glogovac et al. 2016	Generic users	sustainable lifestyles	community	community	User + System	✓	✓	I custom software (smartphone application)	sustainable lifestyles, climate activities, daily challenges	x/✓	✓	Self-Others	Mobile Application	800	SD: 3-weeks	N/A	Device logs
35	Epstein et al. 2016	Generic users	food diaries	peers in the same community (private facebook group)	social network	User + System	✓	✓	I custom software (smartphone application)	complete food challenges	x	✓	Self-Others	Social network (Facebook)	61	SD: 3 weeks	Interview	Device logs, Questionnaires
36	Nicholson et al. 2016	Women who diagnosed GDM	healthy eating	peers	peers in the same conditions	User + System	✓	✓	commercial product (digital electronic scale)	Gestational weight gain (GWG), Laboratory, Self-reported behaviors, Infant measures	✓	✓	Self-Others	Online platform	10-23= 33	SD: 6 weeks	Interview	Device logs
37	Hsu et al. 2016	Patients with type 2 diabetes	medication and blood glucose	healthcare providers (HCPs)	healthcare providers (HCPs)	User + System	✓	-	I commercial hardware (A wireless glucose meter) and custom software (tablet computer application)	medication adherence and blood glucose	x	✓	Self	Online platform	40	SD: 12 ± 2-week	Interview	Device logs, Questionnaires
38	Adami et al. 2016	Generic users	call, SNS, application usage and physical activity trends	selective audience	selective audience (a group of friends)	User + System	✓	✓	I custom software (mobilephone app and online weblogs)	physical activity, calls, SMS, app usage and mobility patterns	✓	✓	Self-Others	Online platform	13	SD: 4 weeks	Interview	Device logs, SNS and call logs, Survey
39	Mertis et al. 2017	Patient	steps	neurologists	2	System	✓	✓	I commercial hardware (Fitbit Zips)	activity	x	✓	Self	F2F Talk	9P+2D UD: 4 weeks	-	Video, Observation	N/A
40	Figueiredo et al. 2017	(Infertile) Women	peer support information, tools (e.g., temperature charts)	Fertility/Infertility/IVF in forum community	forum community	User	✓	✓	I personal choice	personal choice	✓	✓	Self-Others	Online health forum	-	-	Forum database analysis	N/A
41	Mau et al. 2017	Mobility impairments users	automatically tracked health and fitness data	physical occupational or recreational therapists	forum community	System	✓	✓	I commercial website (PatientsLievle)	activity	✓	✓	Self-Others	Online health forum	10Thurs 10P=20	SD: 1 hour	Interview	N/A
42	Chung et al. 2017	Generic users	food	social network	social network	User	✓	✓	I Camera	photos of food	x	✓	Self	Social network (Instagram)	16	SD: 45-77 mins	Interview	N/A
43	Puassaint et al. 2017	Generic users	steps	social network	social network	System	✓	✓	I Commercial software	steps	x	✓	Self-Others (Ranking)	Application	32	SD: 30-120 mins	Interview	N/A
44	Puassaint et al. 2017	Generic users	Heart rate	friends and co-workers	working group (3-5)	System	✓	✓	I commercial hardware (Fitbit)	heart rate data	x	✓	Self-Others	Online platform	18	SD: 15-20 mins, UD: 2 weeks	Interview	N/A
45	Dolegova et al. 2017	Nutrition hobbyists	dietary struggles, personal or intimate information	online soylent user community	community	User	✓	✓	I commercial website	impact of the diet on mood, digestion, bowel movements...	✓	✓	Self-Others	Online platform	43	SD: 11 months	Interview & Observation	N/A
46	Edelen et al. 2017	Generic users	wedding memory	wedding participants	wedding participants	User	✓	✓	I documentary	documentary (photos, time...)	x	-	Self-Others	Brochure	12	-	Interview	N/A
47	Couchbury et al. 2017	Generic users	self-logged food diaries	social network	social network	User + System	✓	✓	I commercial hardware (MyFitnessPal) and I commercial software (Twitter)	Calorie goals, intake and consumption	x	✓	Self-Others	Social network (Twitter)	691	-	N/A	Users' post
48	Larsen et al. 2017	a PTSD-suffering veteran with bowel syndrome (BS) patients	mental health	therapist	2	System	✓	✓	I custom hardware (a wrist-worn snarebust) and a smartphone	a bodily experienced precursor to hyperarousal	x	✓	Self	Application	1	UD: 100 days	N/A	N/A
49	Schroeder et al. 2017	Generic users	food and symptom	providers	2	User + System	✓	✓	I custom platform	patient's food and symptom journal	x	✓	Self	Website	10HCP +10P=20	SD: 1 hour	Interview	N/A
50	Pedrojo et al. 2017	Generic users	mood	colleagues and managers	colleagues	User + System	✓	✓	I self recording	self-reported mood	✓	✓	Self-Others+ AvgOthers	Application	88	17+7= 14 weeks 2UD: 26 days	Interview	1) mood & questionnaires 2) Device logs
51	Kozianus et al. 2017	Children with type one diabetes	blood glucose data	parent-caregivers	parents	User + System	✓	✓	I custom software (Nightcoud) + custom hardware (GGM sensor) + smartphone	blood glucose data	x	-	Self	Application	20	SD: 12 months+	Interview & Observation	N/A
52	Luscio et al. 2017	Students and teachers	emotional shift	teacher (giving lecture)	offline group	User + System	✓	✓	I commercial hardware (Empatica E4 wristband), camera, questionnaires (PAM, ETS, USE)	emotion, recording of a lecture, self-report emotion	-	✓	Self-Others	Mobile Application	54	SD: 36 lectures	N/A	Device logs, Emotion, Questionnaires
53	Lusue et al. 2017	Obesity children	physical activity and food	parents in the same community	social network community (limited group members)	User + System	✓	✓	I commercial hardware (Activity AX3), commercial software (mobile and social media (WhatsApp and Instagram))	their children's food intake	✓	✓	Self-Others	Social network (WhatsApp, Instagram)	50	SD: 6-times camp event/weekend data/ summer break	N/A	Device logs, children's self-reported parents post

(b) Part-2

Fig. 2. (continued).

#	Ref	Users	Sharing Context				Data Collection				Feedback		Evaluation					
			What to share	Share to whom	Group size	Control	Benefit	I/C	Tool	Content	Ano	Vis Me	Format	Platform	No.	Duration	Qualitative	Quantitative
54	Meng et al. 2017	Young adults	Fruit and Vegetable Intake	3 other anonymous strangers (participants)	3 confederates in the same group	User + System	✓	✓	I custom website	post self-tracking messages to report their fruit and vegetable consumption	✓	✓	Self-Others	Online platform	111	SD 4 weeks	N/A	Device logs, Questionnaires
55	Liu et al. 2017	Generic users	Heart rate	family and friends	2	User + System	✓	✓	I commercial hardware (Mio Alpha 2 watch)	Heart rate, activity, messages they sent related to the study	x	✓	Self	Mobile Application	13	SD 2 weeks	Interview	Questionnaires
56	Kim et al. 2017	Clinician	patient-generated data	patients	-	User + System	✓	✓	I -	Activity, meal, sleep, weight, stress, blood pressure	x	-	Others	Shared display	18	SD 15 months	Interview & workshop	Survey
57	Kolbert et al. 2018	Newborn baby	parent-tracked baby data	HCPs (healthcare professionals)	3-4 (family members & one HCP)	User + System	✓	✓	I prototypes (custom hardware (physical and digital platforms/apps, researchers' dashboards))	feeding, sleeping, diapers, milestones..	x	-	Self (Parent-child) + Self (baby data)	Data-enabled cameras	34HCPs +2families	SD 5 weeks	Interview & Design	Device logs
58	Ayala et al. 2018	Generic users	bullet journaling	social network	social network	User + System	✓	✓	I drawing & commercial software (Instagram)	bullet journaling (habit, mood, and symptom tracking)	x	✓	Self-Others	Social network (Instagram)	-	-	Photos & comments on Instagram	N/A
59	Gaibetti et al. 2018	Younger populations	steps	classmates	community	System	✓	✓	I commercial hardware (Xiaomi MiBand 2) and commercial software (Google Fit, Google Maps, Google Messages) and custom website	steps	✓	✓	Self-Others	Shared display	30	UD 6 weeks	Interview & Observation	Device logs
60	Lakoff et al. 2018	Generic users	food journal	family members	family members(2-5)	User + System	✓	✓	I commercial software (WhatsApp)	a photo of your meal with a short description	x	✓	Self-Others	Social network (WhatsApp)	158-31 =189	UD 12 days	Interview,photo, messages	Users' post, survey
61	Festel et al. 2018	Generic users	Daily Physical Activity, Perceived Stress, Digital Social network	people like me	-	System	✓	✓	I custom platform	Daily Physical Activity, Perceived Stress, Digital Social Activity	✓	✓	Self-Others	Online platform	47	SD 3 weeks	Interview	N/A
62	Chen et al. 2018	Stroke patients	stroke rehabilitation experience	peers in the same community	peers+community	User + System	✓	✓	I camera	record log during rehabilitation	✓	✓	Self-Others	Social media (YouTube)	15	-	Analyze logs	N/A
63	Rosko. 2018	Social and psychosocial problem youths	physical activities and sleeping habits	peers in the same community	peers lives in the HVB homes	System	✓	✓	I commercial hardware (Jawbone UP24) and commercial software (accompanying app)	daily physical activity and sleep patterns	x	✓	Self-Others	Mobile Application	8youth+12staff=20	SD 4 month	Interview	Device logs
64	Estrope et al. 2018	Generic users	physical activity	online community (of Fitbit)	forum community	User + System	✓	✓	I commercial hardware (Fitbit) and commercial software (Fitbit forum)	steps throughout the workday	x	✓	Self-Others	Online platform	-	-	Posts analysis	N/A
65	Kuening et al. 2019	Depression / anxiety young adults	physical activity	group members (user created)	patient group (user created)	System	✓	✓	I software (wearable activity tracker Garmin Vivofit and its mobile app)	steps	x	✓	Self-Others+ Ranking	Mobile Application	22	SD 10 weeks	Interview	Fitness level
66	Christinger et al. 2018	Generic users	color-coded road by EDA signals	neighbourhood citizen	community	User + System	✓	✓	I commercial hardware (Empatica E4) and custom software (DT app)	photos, audio narratives, and participant-assigned vignettes for the specific built environment	✓	✓	Self-Others	Mobile Application (smartphone/tablet APP)	14	SD 2 days	Posts analysis	Spatial data and EDA
67	Baranyi et al. 2019	People with diabetes (type1&2)	performance toward a health challenge (experience points XP)	peers in the same community	2, compete with one peer in the same community	User + System	✓	✓	I custom software (web-enabled application)	Time stamp, blood glucose level, carbohydrates consumption, insulin dose, Physical activity, State of mind, Additional notes	✓	✓	Self	Online platform	10	SD 5 weeks	UCD approach, Interview	Device logs, diary entries, game played
68	Wiet et al. 2019	COPD patients	Oxygen Saturation, Pulse, Weight, Blood Pressure, Multiple Choice, Free-form Text	healthcare providers	2	System	✓	✓	I commercial software, two telehealth systems	body data	x	✓	Self	Telehealth systems	6	UD 3-24 months	Interview	N/A
69	Kim et al. 2019	Generic users	qualitative and nuanced aspects of personal data	strangers	2	User + System	✓	✓	I custom website and commercial hardware (Fitbit and Apple Pencil)	sketch	x	✓	Self-Other(1)	Online platform	14	SD 1 hour	Interview	N/A
70	Kim et al. 2019	Autism spectrum disorder (ASD) adolescents	data about self	friends and family, clinicians	3-4 (family members & clinician)	User + System	✓	✓	I custom self-tracking platform	self-decided data type	x	x	Self	F2F	5	2 weeks	Participatory design	N/A
71	Garcia et al. 2019	Generic users	sleep patterns, steps, workouts, nutrition, mood, and weight	acquaintances	acquaintances	User + System	✓	✓	I custom website (online logs) + commercial hardware and commercial software (Jawbone UP 3 app)	diary (fitness goals, exercise, meals, mood) + physiological and activity data including resting heart rate, quality of sleep and steps	x	✓	Self-Others	Online platform	-	SD 6 months	Duethnograph y, diary, dialogue role	Device logs
72	Paldán et al. 2019	Peripheral arterial disease patients	physical activity	community	community	User + System	✓	✓	I custom software (smartphone application)	steps, active minutes, progress, adherence	✓	✓	Self-Others+ Ranking	Mobile Application	45	SD 3 months	Interview	Device logs, Questionnaires
73	LIU et al. 2019	Generic users	biometrics	one chosen partner	2, another chosen partner	User + System	✓	✓	I custom software (smartwatch application)	heart rate	x	✓	Self-Others	Smartwatch Application	34	SD 2 weeks	Interview	Survey
74	Xue et al. 2019	Office workers	stress(HRV)	colleagues	4 colleagues	System	✓	✓	I custom hardware(FPG-Arduino) + software (Processing)	heart rate variability	✓	✓	Self-Others	Shared display	24	SD 1 hour	Interview	Device logs, HRV, Questionnaires
75	Chen et al. 2019	People with healthy eating goals & Patients	food	providers	2	User + System	✓	✓	I custom software (mobile App)	food, symptoms	x	✓	Self	web App	33P+16 D=49	SD 3-4 weeks	Interview	N/A
76	Estélin et al. 2020	Generic users	Personal informatics data	social network	social network	User + System	✓	✓	I custom self-tracking platform	self-decided data type (music, steps, calories, duration, heart rate)	x	✓	Self	Episternal social network	506	-	N/A	Survey
77	Popov et al. 2020	Adolescents	Personal informatics data	peers	4 students	User + System	✓	✓	I custom software (smartphone application)	self-decided data type (diet, health, mood, steps)	✓	✓	Self	Mobile Application	4	2 years	Interview	N/A
78	Sassano et al. 2020	Caregivers (e.g., parents)	Personal informatics data	family (children)	2	User + System	✓	✓	I commercial hardware (MiBand 2) + custom software (MiFit app)	steps	x	✓	Self-Other(1)	Mobile Application	18 families	SD 3 months	Interview	N/A
79	Fonseca et al. 2020	Elders	physical activity	peers	peers in the same community	User + System	✓	✓	I commercial hardware (a pedometer bracelet) + custom software (smartphone application)	steps	x	✓	Self-Others	Online platform	14	SD 2 months	Interviews & focus group	Device logs, steps
80	Wanindien et al. 2020	Parinson's diseases patients (PwP)	health data	Health care professionals (HCPs)	healthcare providers (HCPs)	User + System	✓	✓	I custom software (smartphone application)	most recent medication list and prescriptions	✓	✓	Self (+Expert comments)	Mobile Application	7P-HH CP+37P=10HCP=49	CH-37P=10HCP=49 4.5 days	Workshop & focus group	Questionnaires

(c) Part-3

Fig. 2. (continued).

#	Ref	Users	Sharing Context				Data Collection				Feedback			Evaluation	
			What to share	Share to whom	Group size	Control	Benefit	Tool	Content	Ano	Format	Platform	No.	Duration	Qualitative
81	Khan et al. 2020	Systemic lupus erythematosus patients	symptoms and delay, environmental data	health coach	healthcare providers (HCPs)	User + System	✓ (Self-Others)	I custom software (smartphone application) and custom website	lifestyle activities (e.g. diet, sleep habits, physical activity, bowel movements) and symptoms	x	✓ Self (+Expert comments)	Mobile Application	50	16 weeks	N/A
82	Sportswood et al. 2020	Leisure-time runners	running activities, achievements, efforts	community	community	User	✓ (community)	I custom software (smartphone application)	running activities, achievements, efforts	x	✓ Self-Others+ Ranking	Mobile Application	17	-	Interview
83	Lue et al. 2020	Patients	patient-generated data	clinicians	-	User	✓ (clinicians)	I paper journal, mobile app, website	exercise, sleep, diet, heart rate, blood pressure	x	✓ Self	F2F	109	-	N/A
84	Gordell et al. 2021	Women and their partners who are trying to conceive	fertility information	partner	2	User + System	✓ (Self-Others)	I custom software (mobile App)	fertility-related data (temperature, ovulation predictions)	✓	✓ Self	Mobile Application	30	-	Interview
85	Phonwath et al. 2021	Patients with Chronic Obstructive Pulmonary Disease	vital parameters (pulmonary function, oxygen saturation, heart rate, temperature)	HCPs (healthcare professionals)	-	User + System	✓ (professionals)	I custom platform (Samsung tablet)	body temperature, increased breathlessness, increased coughing, changed color of sputum, heart rate, oxygen saturation, FEV1	✓	✓ Self	Mobile Application	87	1 - 791 days	N/A
86	Pigman et al. 2022	health self-trackers	self-tracked health-related data	research institutes, scientists	-	User + System	✓ (researchers)	I commercial software (Runastic Running App)	personal physical activity	✓	✓ Self	Mobile Application	919	-	N/A
87	Su et al. 2022	Patients with chronic diseases	peer support behavior, task completion, health-related outcomes	patients in the same community	5	User	✓ (Self-Others)	I commercial software (WeChat App)	peer support behavior, task completion, and health-related outcomes	x/y	✓ Self-Others+ Ranking	Mobile Application	25	SD: 3 months	Group chat messages
88	Kyle et al. 2022	Pregnant women diagnosed with Gestational Diabetes Mellitus	glucose levels, activity data, food records	HCPs (healthcare professionals)	-	User	✓	I commercial software (eMOM GDM app) + commercial hardware (activity tracker), continuous glucose meter	glucose levels, activity data, and electronic 3-day food records	x	✓ Self	Mobile Application	200	-	Interview
89	Gowants et al. 2023	Patients with coronary artery disease, valvular disease, arrhythmias	daily physical activity levels, dietary habits, mental stress, smoking, sleep quality	practitioners and relatives	3-5	User	✓ (practitioners)	I custom software (online platform with a custom-built chatbot) + commercial hardware (Philips Health Band)	physical activity, self-reported lifestyle data (daily physical activity levels, dietary habits, mental stress, smoking, sleep quality), and physiological data	x	✓ Self	Online platform	100	1 year	Interview
90	Morgan et al. 2021	Members of online communities related to chronic conditions trying to conceive	Personal self-tracking experiences, emotional reactions to data	online community members	community	User	✓	I commercial website (online communities on Reddit.com) + commercial hardware (wearables and digital tools for self-tracking)	users' personal experiences with self-tracking, emotional reactions to self-tracking data, and interactions with the online communities	✓	✓ Self-Others	Online platform	-	SD: 6 months	Posts analysis
91	Lakmal et al. 2021	Young adults	Drinking behavior, social occasions	research institutes, scientists	-	User + System	✓ (researchers)	I custom software (embedded smartphone sensors, App)	Smartphone sensing data (accelerometer, microphone, geotag, proximity), self-reports from participants	✓	x Self-Others	Mobile Application	241	SD: 3 months	Self-reports
92	Rachael et al. 2021	Teams collaborating on projects or initiatives	Team collaborative activities	researchers and team members	team	User + System	✓ (researchers)	I custom software (web, mobile App)	Team activities (e.g. team meetings, meetings, education/outreach event, planning tasks, brainstorming)	x	✓ Aggregation ALL	Mobile Application	20	SD: 50 days	N/A
93	Xi et al. 2021	Generic users	Tracking data, progress, healthy eating practices	social platforms, peers, community	peers, community	User	✓	I commercial software (food journaling app)	Tracking data, progress, and knowledge related to healthy eating practices	✓	✓ Self-Others	Mobile Application	18	-	Interview
94	Awasetha et al. 2022	Black Americans	Self-tracking data related to heart-healthy activities	Community members, program organizers, health workers, public health advocates	community	User	✓ (community)	I self-chosen tools	heart-healthy activities (e.g. step and food tracking), culturally relevant content	x/y	✓ Self-Others	F2F Talk	15	-	Interview
95	Michael et al. 2022	Patients with various mental illnesses	mental health related data	Clinicians, individuals managing BD, close contacts	community, clinicians, close contacts	User	✓ (clinicians)	I custom software (mobile App)	Affective variables (emotions, mood), behavioral proxies (sleep, exercise) related to mental health	x/y	✓ Self-Others	Online platform	1	-	N/A
96	Ahne et al. 2022	Office workers	collaboration experience	colleagues	colleagues	User	✓ (community)	I notebook	Metrics related to collaboration experience, workflow, collaboration practices and preferences	x/y	✓ Self-Others	Notebook	15	SD: 2 weeks	Interview
97	Dennis et al. 2022	Strapathal users	Personal informatics data	Friends, family	social network	User + System	✓ (Self-Others)	I commercial software (SnapP App)	PI data (steps, heart rate, food time, music)	x	✓ Self	Mobile Application	21	SD: 3 weeks	Posts analysis, interview
98	Romain et al. 2023	High school / higher education students	collaboration practices and experiences	group members	Group size	User	✓	I custom hardware (CoSense), a physical self-tracking kit + custom software	Physical events in the collaboration environment (e.g. noise levels, air quality, temperature, light, sound data)	x/y	✓ Self-Others	Physical self-tracking kit	-	-	N/A
99	Shree et al. 2023	Cardiac rehabilitation patients	activity, experience	clinicians	clinicians	User	✓ (clinicians)	I commercial hardware (Fitbit) + commercial software (Excel)	heart rate (HR), blood pressure, step, step count, experiences	x/y	✓ Self	Online platform	16	SD: 18 weeks	Co-design, interview
100	Mukherjee et al. 2023	Indians	Personal health information/experience	Users with similar health profiles	patient group	User	✓ (therapists)	I custom platform (Cloud-Based PHR and PHN platform)	Personal health records (e.g. blood glucose) self-tracked data on day-to-day health conditions (e.g. eating, exercise)	✓	✓ Self-Others	Online platform	-	-	N/A

(d) Part-4

Fig. 2. (continued).

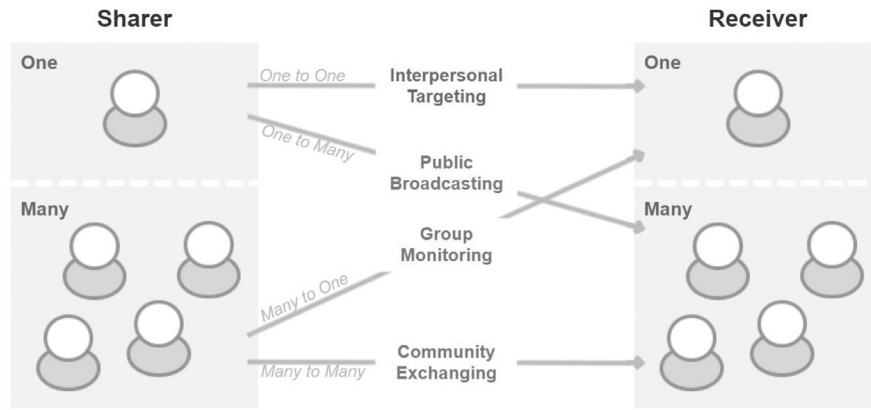


Fig. 3. Four types of sharing according to the sharer-receiver network.

Contextual Aspects			
Sharer	Sharing purpose	Receiver	Sharing content
	Exchange support: give and receive support, community empowerment; Get Information: ask questions, get comments, get feedback from others; Comparison: to better understand self through knowing others; Connectivity: improve connection emotionally; Collaboration: finish something or achieve a goal together; Reward: money incentive.	Family & Friends: personal relationships; Professionals: professional relationship; Peers: co-workers, classmates; Acquaintances: neighborhood; Strangers:	Physical Activity: steps; Food & Nutrition: food intake, calorie intake, nutrition; Body data: heart-related, blood glucose, saliva sample, skin conductivity; Thoughts & Memories: journaling, wedding memories; Stress & Emotion: mood, emotional shift, stress; Sleep: sleep hours, sleep quality; Other Activities: location, habits, personal profile, lifelog, performance.
Sharer	Group size	Sharing platform	
	Generic Users: not specifically targeted users; Specific Groups: - Patients: people who need medical treatment; - Family members: - Students: - Workers: - Special-care groups: babies, children, elders; - Local residents: - Event participants: wedding, lecture, running; - Other specific groups: nutrition hobbyists, smokers.	XL: >20 L: 7-20 M: 5-7 S: 3-5 XS: 2	Mobile-based Applications: custom/commercial App; Online Forum: online community; Co-located Shared Display: co-located shared display; Social Network: Facebook, Twitter, Whatsapp, Instagram; Printed Material: face to face talk; Other Platform: drinks, recording.

Fig. 4. Keywords extractions.

- **Sharer:** We recognize two types of roles of the sharers: generic users (35%) and specific groups (65%). “Generic users” describes the system designed for the general population, without specifying a targeted user group. For instance, generic descriptions like “people” [51], “users” [60], etc. are involved in this subclass. “Specific groups” describes the system designed for a specific group of users. For instance, patients ($n = 27$), family members ($n = 3$), students ($n = 4$), workers ($n = 8$), special-care groups ($n = 8$), local residents ($n = 3$), event participants ($n = 5$), other specific groups ($n = 7$). “Special-care group” explains system designed for users who need special care (e.g., users with mobility impairments [13], not familiar with technology [32], obesity children [61], newborn babies [62], depression young adults [63], etc.). “Event participants” gathers users in co-experienced events (e.g., lecture [64], running [33], social health activities [54], wedding [65]). “Other specific groups” contains users from specific communities (e.g., smokers [30], single people [25], long distance couples [66], nutrition hobbyists [67], etc.). *Public broadcasting* systems are mostly designed for generic users (80%). Whereas, *Interpersonal targeting* (76%), *Community exchanging* (62%), and *Group monitoring* (100%) systems are commonly designed for specific users. Among which, *Group monitoring* contained by patients (78%), special-care user group (11%), and other specific groups (11%) only (Fig. 5b).
- **Receiver:** We classify five groups of receivers according to their relationships with the sharers: family & friends ($n = 31$), professionals ($n = 21$), peers ($n = 41$), acquaintances ($n = 4$), and strangers ($n = 3$). “Family & friends” contains family members [68,69], partners [26,56], friends [55,59], as well as friends on social network [70,71]. Medical experts [30], HCPs [49],

professionals [29] are coded as “professionals” group to describe the professional relationship between the receiver and the sharer. “Peer” includes co-located peers (e.g., co-workers [27,72], classmates [14]) as well as peers from online communities (e.g., infertility forum [12], PatientsLikeMe [13]). “Acquaintances” contains local neighborhoods [52,73,74] and acquaintances [18]. *Interpersonal targeting* commonly happens between people with relatively close personal (44%) or professional (44%) relationships, which echoes with the microlayer relationship in Murnane et al.’s ecological model [5]. In *Public broadcasting* category, the receivers’ role moves from microlayer to exolayer [5]. In *Group monitoring* category, receivers are all HCPs (66.7%) and medical experts (33.3%), who share a professional relationship with the sharers. In *Community exchanging* category, receivers spread from people who are closely related with (family & friends, peers) to acquaintances (local neighborhoods) (Fig. 5c).

- **Group size:** We classify the group size according to the number of users included in the sharing activity: XL (group size > 20, e.g., online community), L (7–20, e.g., classmates), M (5–7, e.g., running group), S (3–5, e.g., family members), XS (2, e.g., partner). Most commonly, the sharing happens either with a large number of users online (46%) or with a small group of users: 2 (25%) members, 3–5 (18%) members. Middle-sized (7%) and large group (4%) have a relatively smaller share among all the reviewed cases (Fig. 5d).
- **Sharing content:** The most popular sharing contents are: physical activity (26%), food & nutrition (13%), body data (15%), thoughts & memories (8%), stress & emotion (6%). “Body data” consists of relatively high-private data types (e.g., heart-related

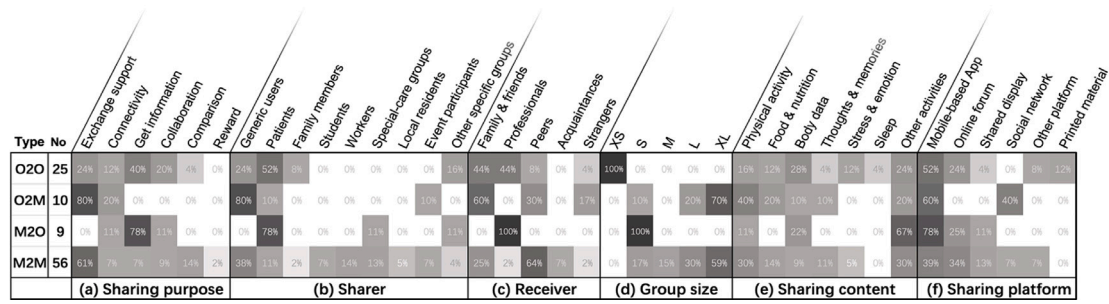


Fig. 5. Contextual aspects of the four types.

[56], blood glucose [75], Oxygen saturation, and lung function [76]). “Other activities” collects relatively less-popular sharing items (e.g., location [77], smoking habit [30]). Some systems share more than one item across subcategories, in this case, it is arranged under the “other activities” subcategory. *Interpersonal targeting* and *Community exchanging* relatively share diverse data content with different privacy levels. *Public broadcasting* often shares not-so-private data (e.g., physical activity, food) in a spreading fashion. Conversely, *Group monitoring* usually shares highly private data types (e.g., body data, medical records) to professionals under sharers’ consent (Fig. 5e).

- *Sharing platform*: Addressing the way that the sharers spread their data to the receivers, we classify the sharing platform into 6 sub-classes: mobile-based applications (48%), online forum (26%), co-located shared display (9%), social network (8%), printed material (3%), and other platforms (6%). “Other platform” involves media other than visualizations (e.g., drinks [26], recording clips [52]). *Interpersonal targeting* and *Community exchanging* share the most variable sharing platforms compared with the other two types of sharing. Face-to-face sharing through printed material cases are all found in the *Interpersonal targeting* category because the format of paper-based sharing is not easily spread. However, it can be efficiently discussed in detail within small groups. The sharing platform of *Public broadcasting* normally happens online to support efficient message delivery. The platform of *Group monitoring* commonly uses customized or ad-hoc applications to satisfy specific needs (Fig. 5f).

3.3.2. Summarize the trend

We map all the contextual factors along the time dimension to discover the potential trend and opportunities of sharing (Figs. 6, 7). There is an overall increase of shared PI cases with time, which indicates an increasing research interest in social sharing in personal informatics. The main *Sharing purpose* are: exchange support, get information, and comparison for a better understanding of self (Fig. 6a). Addressing the *Sharer*, researchers’ interest in designing for generic users stays flat in the past decade. By contrast, as can be seen in Fig. 6b, there is an increasing interest in shared PI design for specific groups of users (e.g., patients, workers, special-care groups). The involvement of patients and special-care groups of users reveals a potential possibility of leveraging self-tracking techniques in the clinical field (Fig. 6b). As for *Receiver*, researchers place great emphasis on sharing with relatively close personal relationships (i.e., family & friends) as well as peers within the same community. The emergence of professionals as receivers echoes with the patients sharers (Fig. 6c). The *Group size* of the sharing activities mostly involve users from online groups (XL) or small groups (XS, S). Middle-sized (M) sharing and large-sized groups (L) are relatively less addressed (Fig. 7d). The *Sharing content* covers wider aspects of people’s lives. Other than physical activity and food, body data, stress & emotion has gained increasing research focus over time (Fig. 7e). The research interest addressing PI *Sharing platform* is increasing in the channel of mobile-based applications and online

forums (Fig. 7f), which mainly depends on the sharing requirements. For instance, when the receivers move from acquaintances to strangers online (Fig. 6c), sharing platform will change accordingly (Fig. 7f).

An overview of the sharing flow is visualized in the Sankey diagram (Fig. 8). In summary, this section introduces six contextual aspects of shared PI (*Sharing purpose*, *Sharer*, *Receiver*, *Group size*, *Sharing content*, *Sharing platform*). We cluster the four types of shared PI addressing these contextual factors to identify the differences and connections. And we find, *Interpersonal targeting* (O2O) often describes the data sharing activity between two individuals with a relatively close relationship. *Public broadcasting* (O2M) often describes low-private data sharing from one individual to a wide range of audiences in a mono-directional fashion. *Group monitoring* (M2O) often describes a specific user group passively share their high-private data to limited numbers of professionals. *Community exchanging* (M2M) often describes a group of users with similar interests shares for reciprocal purposes. Moreover, we map the contextual aspects along the time dimension and summarize the trend of current research interest in the social sharing of personal informatics. These contextual aspects often appear when users decide what tool to use to track what data, what will be shared for what purpose, who to share with, etc. These questions formed and correspond to the “preparation stage” in the classical PI model [1].

3.4. Data collection of shared PI

Tools for data collection in shared PI practices are not limited to use PI tools, which means the initial purposes of inventing these tools are not necessarily designed for PI practice. For example, people use commercial systems like social media [51] as data collection tools for food-tracking in social PI practice. Understanding how people use these techniques to collect data and social aspects in PI practice can facilitate future designers and researchers to identify key areas (e.g., what tool can be used in specific PI data collection, what roles can be involved in data collection process) for data collection in future sharing practice. Therefore, in this subsection, we will put our emphasis on the data collection tools, techniques as well as how social aspects affect the data collection process.

3.4.1. Techniques of data collection

There are three ways for data collection addressing different levels of automation in shared PI practices: self-report, mixed sensing (semi-automatic), and automatic sensing. Regarding current collection tools that applied in these shared PI practices, 23% cases use self-report, 44% systems use automatic sensing devices, and the remaining 33% use mixed methods to combine sensor data with self-report data to enrich the context of the collection. The selection of data collection tools depends on the data type (Fig. 9). Automatic sensing devices can efficiently and automatically facilitate the self-tracking process. Steps [78], locations [77], heart rate [23], skin conductivity [79], and blood glucose [49] become accessible anytime and anywhere by the increasing sensing devices. However, technologies cannot cover every aspect of life, we should leverage technologies with the human in the

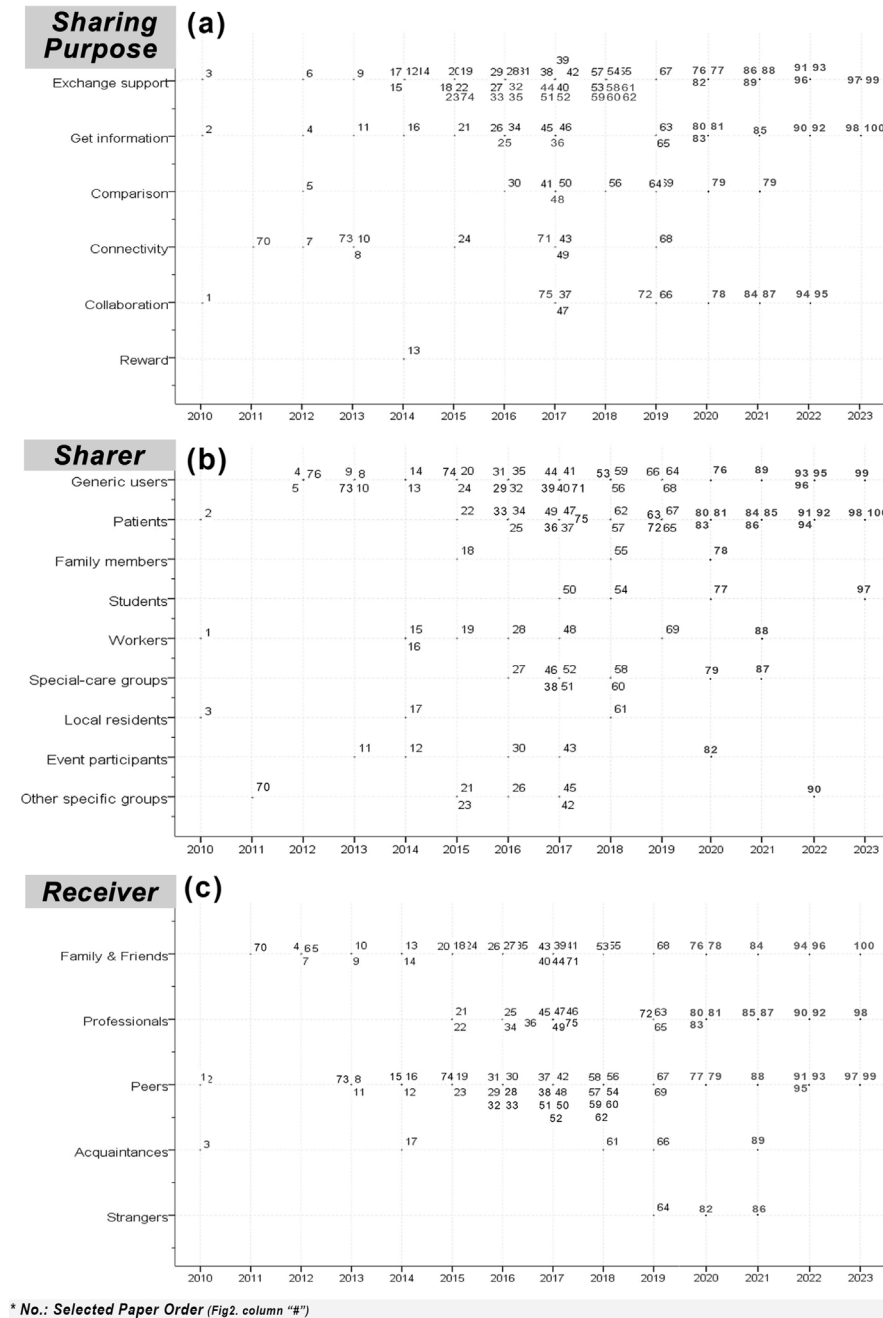


Fig. 6. Contextual aspects change with time (part 1).

loop [80]. Some data collection still needs users' input (e.g., food [81]) or partially rely on self-report (e.g., emotion [82], stress [31]). Manual efforts in data collection can in some cases reassure the accuracy of data input and enrich the context. For instance, sensing technologies can measure physiological emotions, and self-report reflect perceived emotional status [82]. It is especially important in the clinical use of shared PI practices, the shared data can be crucial to diagnose diseases and give feedback. Patients' input can largely inform doctors about their disease and symptom data. Therefore, self-report can complement the technical deficiency to some extent and enrich the sensing context. Sensing technology and self-report are two complementary tools, and the value of each depends on the application scenario and the situation at hand.

Besides the technical deficiency in accurate-sensing, a lot of system design deliberately leaves space for users to participate in and contribute to the data collection process. Although automatic sensing saves

users' time and efforts to collect data, considering users' autonomy and self-efficacy, users' input and participation are still necessary for some of the collection processes. For instance, a lot of shared PI systems involve users to add free-form text with the sensing data (e.g., [76,83]). On the one hand, it makes the shared content more understandable to the receiver(s). On the other hand, it enables users to gain self-efficacy and autonomy in the data collection process.

3.4.2. Shared data collection in shared PI

Shared data collection involves multiple stakeholders, it drives concerns on the data sharing content. The shared collection process can be operated by another user [61]. In this case, the sharers are often people who need special care and are not capable of operating the data-tracking by themselves alone. For instance, the elderly sometimes do not know how to use advanced sensing devices for health tracking, so an intermediary (his daughter) takes care of data collection for

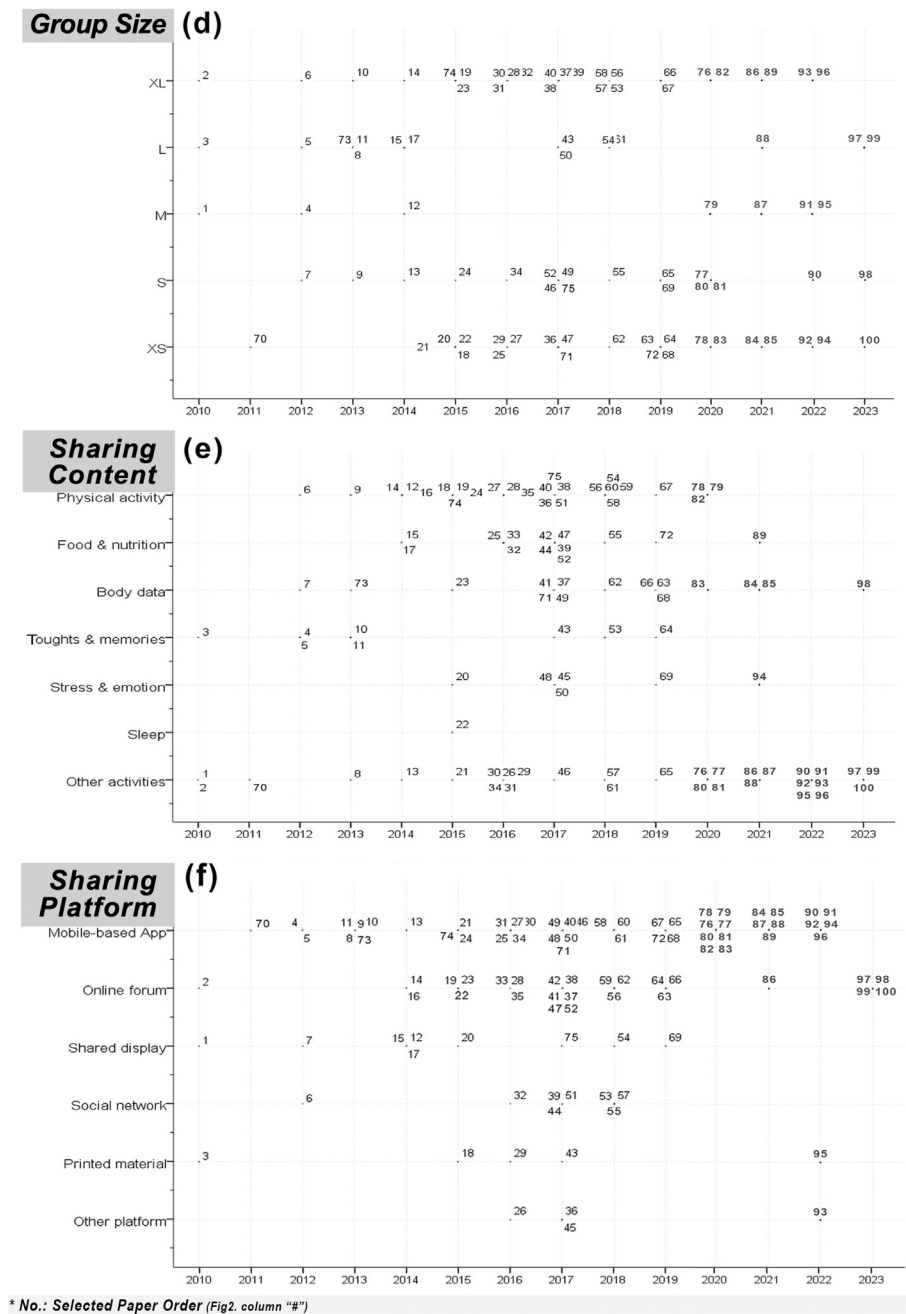


Fig. 7. Contextual aspects change with time (part 2).

him [32]; parents collect data to track the activities of their newborn babies and share it with HCPs [62]. Another type of shared collection is accomplished together by multiple stakeholders simultaneously [31,56] or remotely [12,13]. This type of shared collection often gathered people pursuing a similar goal (e.g., eat healthfully, be more active, less stressed). Shared collection with multiple stakeholders made the tracking more effective [84], made the collection more engaging [82, 85], and made the collected data more carefully-prepared. Because people will consider “their imagined audience in mind” [74]. Though the initial engagement may disappear when people knew what to expect [56] after a few practices.

3.4.3. Summary

We described the shared collection in this section addressing: the techniques used for data collection (self-report, mixed sensing, automatic sensing), the automation and autonomy of the data collection

process, as well as the data collection under social contexts. Overall, tools that be used for data collection are not necessarily designed to collect PI data. Sensing technologies and self-report are two complementary tools in the data collection stage. Current sensing cannot cover every aspect of our needs, and users’ autonomy should be reserved to some extent. Therefore, the participation of users in data collection cannot be replaced. Moreover, the presence of others can make data collection more effective and more engaging. These social aspects can be leveraged to create opportunities for sustainable collection. These aspects formed and correspond to the “collection stage” in the classical PI model [1].

3.5. Media of shared PI

In the classical PI model, the integration stage exists between collection and reflection, and it describes the collected information “be

There are three approaches to achieve privacy protection in current shared PI cases: (1) anonymity; (2) ambiguity; and (3) aggregation. Anonymity is a direct way for privacy protection, it is widely used in offline sharing [14,31], as well as online sharing [12]. Either users themselves take control of the anonymity level [83,88] or the system is designed with an untraceable identity [27,31]. Anonymity is widely applied and is applicable for most sharing contexts. Ambiguity makes abstract the sharing either from the source (data) or from the end (format). When the system designed for interaction and entertainment over precise number-delivering, the collected data can be ambiguity from the source. For example, Moodlight [79] uses an ambient lighting system to represent emotion ambiguously; social HUE supports the abstract share of physical activity data to create social connectivity between the elderly and their caregivers [89]. When users need to access the results clearly and precisely by themselves, the ambiguity may happen at the tail before sharing. For example, DiaBeaThis system obfuscates blood glucose data into fuels and embeds it in a social game [90]. Aggregation is commonly applied when the aggregated result is meaningful or interests the users. For instance, users may need an aggregated result to understand the overall performance of a running group [33] or a community [27,74,81], to compete with other communities as an entity [91], or to view and interact with a cooperated activity [79]. Though, as can be noticed in the paper list (Fig. 2), not-aggregated presentations (visible self and others in the group) stay the most popular feedback content in these shared PI practices. Noticeably, some scenarios concern less about privacy protection under specific conditions. Either the shared data has low-private level (e.g., steps [33,78], open-shared memory [92]), or the receivers are trusted people (e.g., partner [26], friends [79], HCPs [76], co-workers [88]).

3.5.4. Summary

We described the feedback of shared PI in this section addressing: feedback forms, timing as well as privacy concerns. The feedback format depends on the resolution of the delivered information, visualization shares the majority part than other formats in current shared PI cases. The feedback is commonly made approachable to the users and the delivery time is often driven by users' needs (real-time, daily, weekly, etc.). Meanwhile, not all the collected data can be shared. Users should take control of what to share and what not. Three ways can be leveraged to tackle privacy concerns in current shared PI cases: anonymity, ambiguity, and aggregation. The aspects of feedback forms, delivery time, and privacy concerns discussed in this section correspond to the "integration stage" in the classical PI model [1].

3.6. Evaluation of shared PI

We categorize the evaluation of the shared PI practices according to the evaluation goal and approach. Table 3 describes the goals and approaches to evaluation in the literature. 22% of the cases were evaluated to understand the interactions among users (coded as "UU"). For instance, evaluate to understand the interpersonal interactions between patients and clinicians [4,57]; to understand users' data-sharing behavior [25] and sharing motivations [74]; or to understand events [72] or group effects [84] on users. Among these, a majority part of the evaluation (68.2%) deploys qualitative methods to understand users' social behaviors and collaborations. 34% of cases emphasize "Users-System interaction", coded as "US". For instance, evaluate how users interact with the system [32]; how the system affect users' awareness [26,79], reflection [30,31,93], modified behaviors [85,88], social interactions and group performance [33]. The evaluation method is relatively diverse and commonly accompanied by semi-structured interviews or surveys. 3% of cases focus on "System" evaluation (coded as "S"). Namely, to evaluate system performance [94], system usability [95], and system feasibility [86]. Cases to evaluate system acceptance and usefulness are often accompanied by user experiences [64,83]. 20%

studies focus on evaluating interpersonal interaction and user-system interaction (coded as "UU + US"). And 17% cases focus on user-system interaction and system evaluation (coded as "US + S"). Four papers do not mention specific evaluation methods [87,94,96,97].

Previous research either evaluates the work as a PI system or a PI practice. It is rarely seen works emphasize evaluating the effects of sharing practices in each stage of PI. Echo with Rooksby et al. [39], the evaluation of a shared PI is beyond the measurement of "behavior change on gained insights", rather, it is also likely to focus on the gaining of shared understanding and social interpreting. For instance, more reflection [30], increased awareness [26,79], improved motivation [33], engaged logging [85], social collaboration [62] are commonly addressed in these shared PI cases. As for the evaluation approach, qualitative research methods can facilitate researchers to understand why and how questions, and are mostly addressed to understand users or users' experiences of the design. Quantitative analysis is mostly applied through collecting users' usage or interaction through device logs, mostly addressed to evaluate the system. By understanding the effects of sharing on each stage of PI from various cases, it is beneficial to develop evaluation approach that applied specifically for shared PI practice.

4. Discussion

This paper reviews current shared PI practices generated over the past 13 years. We characterize diverse shared PI cases in light of the classical model of PI stages (preparation, collection, integration, reflection, action) [1]. In this section, we discuss some major insights into "shared PI" gained from this review. Moreover, we summarize the design opportunities based on our findings, followed by implications for future research in designing shared PI systems.

4.1. There is more to shared PI than the sharing of data

Personal informatics (PI) is "a class of tools that help people collect personally relevant information for the purpose of self-reflection and gaining self-knowledge" [1]. As shown in our review, an emerging number of studies set out to explore the social aspects of PI. Overwhelmingly, current explorations involving social aspects of PI cases have been focused on the sharing of data visualization or sense-making activities among people, i.e., in the stage of integration and reflection. Nonetheless, by classifying and extracting a diverse set of shared PI cases, we also find that the act of social sharing of PI does not only concern the data display (integration) or sense-making (reflection), but also concerns other stages of PI, such as the stage of preparation [32, 49,57,62,88], collection [33,56,79,85], and action [33,54,76,84,85,91, 106]. And in each of the PI stages, the act of social sharing can be a "double edged" sword that implies both benefits and risks, and thereby needs to be carefully designed for.

Therefore, a key insight generated from our review is that designing shared PI is beyond merely designing shared data visualization, or shared sense-making activities. Rather, it can also mean designing shared preparation (e.g., the targeted user is not familiar with new technologies, an intermediary help to prepare the tracking and support the target beneficiary to maintain health [32]), shared collection (e.g., tools to support group fitness are developed and deployed to collect real-time performance data for running groups [33]), or shared action (e.g., specific group activities (walk together after lunch) were found in an employer-sponsored health program [91]). And this suggests a broader and exciting space for future design to explore and facilitate different acts of social sharing throughout the PI stages. As such, from our understanding, shared PI practices could be described as follows: multiple stakeholders use PI tools to collect personal- or collective-relevant information for the purpose of *Interpersonal targeting* (O2O), *Public broadcasting* (O2M), *Group monitoring* (M2O), or *Community exchanging* (M2M), with the social sharing aspects concerning different

Table 3
Evaluation goal and approach.

Evaluation	Approach\Goal	UserUser	UU+US	UserSystem	US+S	System
Quantitative	Device log + survey/questionnaire	[59,98]	[99]	[82,100]	[77,81]	
	Device log + sensor data			[61]		
	Survey/questionnaire	[101]	[84,102]	[17,88,103]	[29,104]	
	Device log		[105]	[60,106]		[107]
Quantitative+Qualitative	Other material + device log	[108]	[109]	[18,73]		
	Interview + sensor data			[63]		
	Interview + survey	[110]	[32,68,75,111–114]	[27,33,49,55,70]	[64,83]	
	Interview + device log + survey			[14,20,31]	[30,93,115–119]	
Qualitative	Interview + device log	[85,120]		[54,69,121]	[66,90,122]	
	Interview + other material	[25,52,58,62]		[26,67,79,123]		
	Other material (photo, video, observation, etc.)	[4,12,124,125]		[47,126]		
	Interview	[13,42,53,72,91,127,128]	[51,56,57,65,74,129–131]	[23,48,71,76,78,132]	[133]	[86,95]

stages of PI, including preparation, collection, integration, reflection or action. Understanding shared PI practice would be beneficial for future researchers and designers interested in sharing practices in PI studies and discovering new research opportunities.

Meanwhile, to be noted, since sharing act may exist across the PI stages, there could be ethical concerns such as privacy, social translucence [134], or data security in other stages as well than only in the stages of integration and reflection (in which multiple stakeholders share visualizations or sensemaking activities). This means for each stage, the effects and risks of “social sharing” should be thought through respectively, in light of the specific use case and user needs in that stage. And it is highly likely that in one stage sharing acts (e.g., collective activity or social translucence) is beneficial and preferable, but in another stage, it should be moderated or avoided due to specific reasons. For example, a group of students collect their physical activity together, but reflect on the data through pseudonymous avatars for privacy protection [14] (concerns on reflection stage). Similarly, in another example, (infertile) women collect their data (e.g., body temperature) privately and remotely, but reflect together by acquiring peer support or offering help through the community forum online [12] (concerns on collection stage).

4.2. Potential opportunities for shared PI research

From the literature review, personal informatics is getting more and more adopted among the general public with the development of ubiquitous sensing techniques. Research interests increasingly emphasize the sharing of diverse facets of life for various purposes via PI systems. Hence a trend can be seen that social sharing has become an important property in many emerging PI designs. In addition to the findings discussed above, here we identify the remaining gaps and design opportunities with the aim of informing future research.

4.2.1. Opportunity 1: Leverage the benefits and minimize the risks in social comparison

Based on our results, sharing is mostly driven by exchanging support, get information, and comparison. And we notice that there is an increasing emergence in comparison-aimed sharing, in which users are enabled to compare their own data with others [23]. It rooted in social comparison literature in the psychology field [135]. On the one hand, social comparison in PI context can be sometimes stressful and burdensome, for example, if one is left behind by peers. On the other hand, it can assure users by implying the “normal” pattern or “overall” distribution of others, or motivate users to achieve more by displaying others’ progress. Among our reviewed designs, the comparison-aimed cases mostly target generic users [20], students [82], workers [27], event participants [54] sharing with family, friends, and peers. Comparison in patients and special-care groups are relatively rare, which suggests both challenges and opportunities to explore proper and beneficial social comparison for these users. In general, to leverage the benefits and minimize the risks in comparison under

PI context, designers can design the comparison between users with similar conditions. For example, in the context of mobility-impairment users’ activity tracking, participants show a great desire to compare with peers with similar mobility impairments to discover new possibilities in rehabilitation [13]. When the situation is relatively comparable, the experiences from peers would show more value to learn from. Meanwhile, the relatively fair comparison would not make users feel anxious about being left behind too much by peers. Additionally, the privacy protection should be embedded in design to minimize risks through (semi-) anonymous sharing [31], social avatar design [14], or thoughtfully design nuanced levels of ambiguity in the interface, to mediate between privacy and social translucence.

4.2.2. Opportunity 2: Tailor shared PI design to specific user group and ad-hoc scenarios

Our analysis reveals an increasing focus on tailoring shared PI to specific user groups and their unique contexts, moving beyond generic use cases. This trend underscores a shift from early adopters, such as quantified-selfers [2], toward broader demographics, including less tech-savvy users and professional or special-care domains. For instance, our findings indicate that shared PI is being integrated into highly contextualized scenarios such as clinics [76], workplaces [88], classrooms [14], and family settings [69], signaling a growing need for domain-specific solutions. This evolution calls for a deeper understanding of the nuanced needs, routines, and social dynamics of targeted user groups within their ad-hoc environments. Rather than applying generic design principles, researchers and practitioners should leverage insights from our results to explore how shared PI can adapt to the specific workflows, cultural norms, and collaborative patterns of distinct populations. For example, shared PI in a classroom might emphasize engagement through gamified activity tracking to foster peer interaction [14], while workplace applications might prioritize social negotiations to combat sedentary behavior [72]. To advance this area, designers should aim to create context-sensitive solutions that not only enhance users’ engagement but also mitigate potential negative impacts on existing routines and social structures. By grounding the design process in the unique attributes of specific scenarios, this approach paves the way for more impactful and ethically mindful applications of shared PI.

4.2.3. Opportunity 3: Ask first what sharers expect from sharing before whom they want to share with

Our characterization of the four sharing types identified the receivers’ roles by the relationship with the sharer. Echoing with [71], receivers can either be strangers or from the user’s existing social networks, and sharers are expecting differently from different roles of receivers. Combined with our findings, O2O sharers expect to get support, comments, or stimuli from family and friends; O2M sharers expect to get interactions, “views”, or “likes” from public media; M2O sharers expect professional comments, feedback, or suggestions from medical experts; M2M sharers expect community peers’ to share back.

However, the users' "expected receivers" may differ from what they really shared with. For instance, when we ask users to whom they would like to share their personal data, they are highly likely not going to say "strangers". Whereas, a lot of previous research had stated, to some extent, people tend to share with strangers with similar conditions or experiences rather than family or friends [71]. One example shows that sharing personal intimate information (e.g., infertility) to strangers through the online community makes infertile women feel more comfortable and secured than sharing with friends and colleagues [12]. Therefore, instead of asking "to whom you want to share" questions, designers should ask (and investigate in-depth) "what do you expect from sharing". In other words, designers can turn the way around by understanding sharers' latent expectations and desires first, then design to catch the "perfect" receivers.

4.2.4. Opportunity 4: Further explore shared PI for co-located groups

Based on the trend of sharing, current research interests in shared PI put great attention on the remote sharing (e.g., share with peers from online forums). However, the design of shared PI for co-located groups (e.g., co-workers [88], classmates [82], lecture audiences [64], wedding participants [65], local neighborhood [74], etc.) is relatively rare. We see an opportunity to design shared PI for co-located user groups under their specific co-experienced context (e.g., occupational stress, lecture presentations, wedding experiences, marathon races, neighborhood events, etc.). Co-located sharing or physically sharing information involves the presence of others, which can foster immediacy and intimacy [136], it creates a sense of interpersonal connectedness and has some unique affective benefits that are less likely afforded in remote sharing [137]. Co-located shared PI can engage in conversations and co-interpretations, it can be inspiring, stimulating, and engaging in sociability. However, not all interpersonal interactions are welcoming and engaging. Accordingly, designers should beware of the co-located interpersonal dynamics (e.g. comparisons [84], social pressures, impression management concerns [138], biased judgments [31], awkward [139]), and the negative emotional outcomes (e.g., stressed, guilty, embarrassed [140,141]) in design for the co-located user groups.

Therefore, while designing for co-located PI suggests many unaddressed opportunities for future designers, it also poses new challenges regarding social acceptance and willingness. Designers are tasked to resolve the tension between these benefits and the concerns. For instance, designers can emphasize the benefits of sharing to targeted users by inviting their trusted parties (e.g., managers to employees, teachers to students) to introduce or to take part in the design process. Such as, inviting managers to participate in the design of mood-sharing tools in the workplace for their staff engage acceptance of self-tracking and benefits both parties under their existing structures (e.g., team cohesion and empowerment of employees) [27]. Additionally, designers may need to caution the inadvertent, unintended self-revelation and the risk of automated disclosure [79] in co-located sharing. Designers can have more nuanced solutions for privacy protection or mediation by navigating the approaches that we summarized in our review: anonymity, ambiguity, and aggregation. For example, designing synchronized activity tracking using pseudonymous avatars for co-located students in the classroom can afford a more public discussion without comprising their privacy [14]. Designers can also leverage game design elements in these non-game contexts (gamification [142]) to ease the sharing and address certain burdensome facts in a fun and playful way.

4.2.5. Opportunity 5: Addressing asymmetrical needs of sharers and receivers in shared PI systems

Our analysis highlights the critical need to design shared PI systems that address the asymmetrical needs of sharers and receivers. This asymmetry becomes especially evident in scenarios where stakeholders have distinct roles, such as patients and doctors. For instance, while patients might require detailed tracking data, including progress reports and graphs for self-reflection, doctors may find such granular

information overwhelming and instead benefit from succinct overviews to support clinical decision-making. Although previous work has emphasized sharers' needs, our findings reveal a broader opportunity to design systems that balance these asymmetrical requirements across diverse contexts, not limited to healthcare. For example, shared PI systems in workplace collaborations or family health tracking similarly require nuanced understanding and design to accommodate differing priorities and interaction patterns among stakeholders. Our review underscores the importance of engaging both sharers and receivers during the design process to uncover shared and diverging needs [21]. By incorporating co-design practices and iterative feedback loops [4], researchers and practitioners can ensure that shared PI systems are better aligned with the goals of all users. This approach not only enhances system usability but also ensures equitable support for all parties, paving the way for more inclusive and effective shared PI applications across varied domains.

4.2.6. Opportunity 6: Lower the sharing barriers by adding options for fine-grained, selective sharing

Users show hesitation about sharing because they may worry their sharing spamming others [71,143,144]. Moreover, the sharing content may disclose the contextual information that users intend to kept private [42]. These concerns hold back on users' sharing activities and thus prevent them from experiencing the benefits of sharing [71]. Therefore, we see an opportunity in designing worrisome sharing to lower the sharing barriers and make the sharing more likely to deploy. To not worried about spamming others, Epstein et al. suggest developing the system to facilitate users to avoid oversharing [42]. To avoid concerns about disclosing contextual information, designers can empower sharers with the ability to selectively share information and control the level of granularity [42]. In other words, sharers should be able to freely select specific people for sharing or able to flexibly editing data before sharing. Designers should keep sharers' autonomy in worrisome sharing. However, to be noted, while enabling such fine-grained, selective controls could enhance user autonomy and reduce their worries, these controls should be provided as additional options and should not be overburdening the users. In other words, users should still be able to share their data via minimal steps in an effortless way. As a result, here designers are tasked with the challenge of balancing user autonomy with the system's ease of use.

4.3. Limitations

This literature review has some limitations that need to be mentioned. First, we only reviewed the scientific literature in the three major databases for literature search. Although they are the predominant sources for the targeted research domain, and we did involve references in these papers to identify potential external papers, it is still possible that we have excluded some related publications that are not covered by these databases. Second, by restricting our search on shared PI practices in scientific literature, we may have missed relevant systems that are not been published but are used in practice, these systems should be studied in future research. Third, given the nature of qualitative analysis, the categorization and characterization of the reviewed design cases might have been influenced by the coders' subjective bias. Although cross review and cross revision between coders and external researchers were conducted to neutralize the potential subjective bias, it is still noteworthy that the categorization and characterization in this work should remain open for future improvement or debate.

5. Conclusion

This paper reviews the existing work on shared PI over the recent 13 years, aimed for analyzing and characterizing current cases to inform future design and development. We identified the different types of sharing and characterized them by sharing purposes: *Interpersonal*

targeting (O2O), Public broadcasting (O2M), Group monitoring (M2O), Community exchanging (M2M). We analyzed how contextual factors differ in different types. O2O type often describes a sharing activity between two individuals with relatively close relationships. O2M type often describes a mono-directional sharing activity without necessarily hoping others to share back. M2O type often describes a specific user group passively share their high-private data to limited numbers of professionals. M2M type often describes a reciprocal sharing activity within an ad-hoc community where people have similar goals or interests. We also examined the feedback design and system evaluation of the existing studies. Based on our review, we argue that the definition of shared PI should be evolved beyond the sharing of data. Rather, it should concern all stages of PI, including preparation, collection, integration, reflection, and action. Along all these stages, we address design opportunities for designers and researchers to navigate the benefits and risks of design social sharing in personal informatics. In doing so, we aim to contribute a set of design-oriented knowledge for the community to jointly advance the research and design of shared PI systems.

CRedit authorship contribution statement

Mengru Xue: Writing – original draft, Methodology, Investigation, Funding acquisition, Conceptualization. **Pengcheng An:** Writing – original draft, Validation, Investigation, Formal analysis. **Zengrong Guo:** Validation, Investigation. **Rong-Hao Liang:** Writing – review & editing, Supervision, Resources. **Jun Hu:** Writing – review & editing, Supervision, Funding acquisition. **Preben Hansen:** Visualization. **Loe Feijs:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank Bin Yu and Kadian Davis-Owusu for their insightful suggestions. This work was supported by Ningbo Innovation Center, Zhejiang University (1140557B20220120), and the China Scholarship Council.

Data availability

Data will be made available on request.

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